

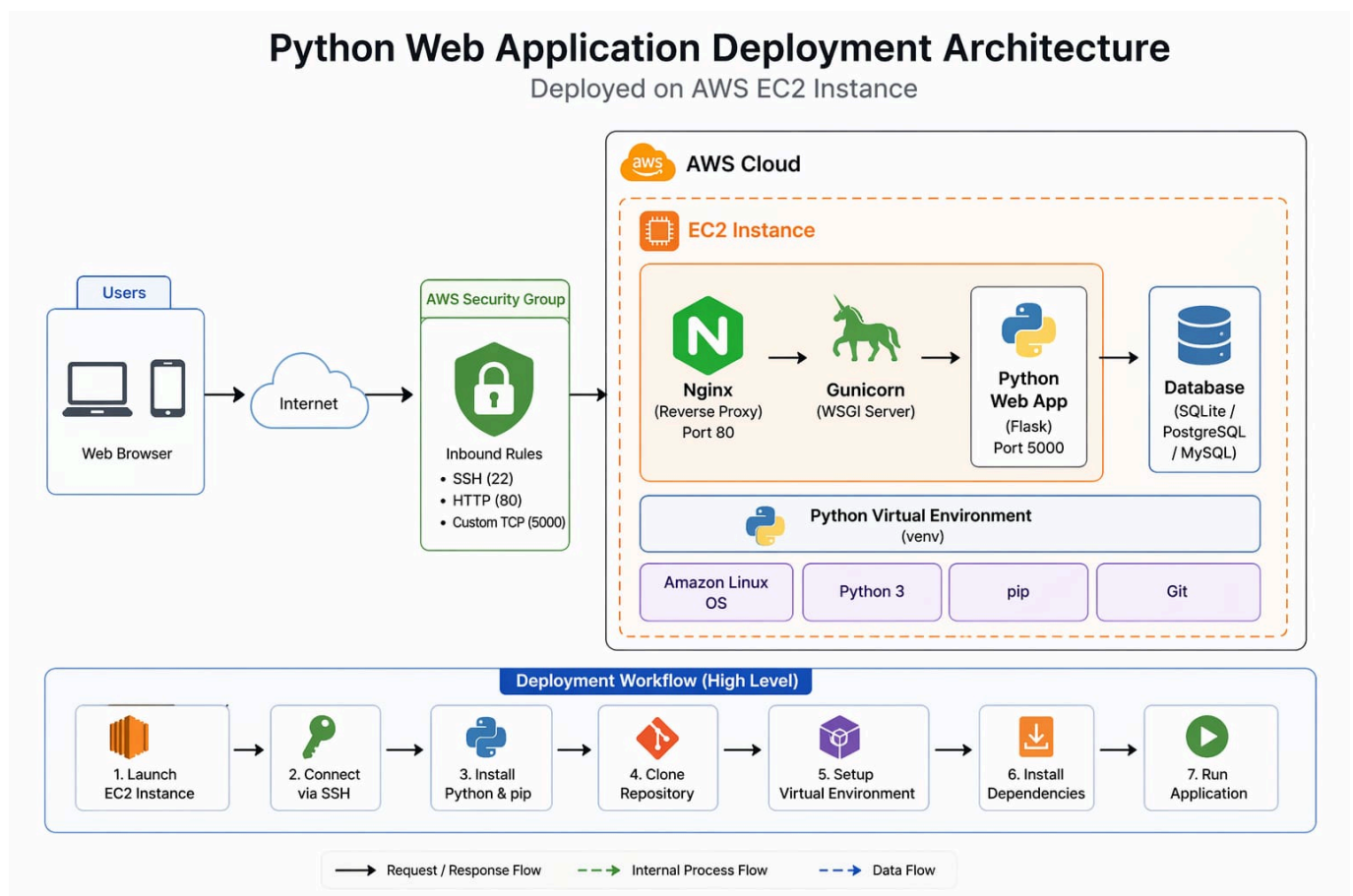
Python Web Application Deployment on Ec2

Introduction

This project demonstrates how to deploy a Python web application on a cloud server using Amazon EC2. It covers the complete process from launching a virtual server to running the application successfully.

The goal of this project is to gain hands-on experience with cloud computing, Linux commands, and Python environment setup.

Architecture Diagram



Steps I Followed

1. launching an ec2 instance

 ec2-user@python:~

3.Installation of PYTHON

```
sudo yum install python3 -y
```

ec2-user@python:~

```
[ec2-user@python ~]$ sudo yum install python3 -y
Last metadata expiration check: 1 day, 0:00:09 ago on Wed Apr 22 18:18:36 2026.
Package python3-3.9.25-1.amzn2023.0.4.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@python ~]$ |
```

- verify version:
python3 --version

4.Installation of pip

sudo yum install python3-pip -y

```
[ec2-user@python ~]$ sudo yum install python3-pip -y
Last metadata expiration check: 1 day, 0:01:02 ago on Wed Apr 22 18:18:36 2026.
Package python3-pip-21.3.1-2.amzn2023.0.16.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@python ~]$ |
```

- Verify pip:
pip3 --version

5.Clone the Repository

git clone url

```
[ec2-user@python ~]$ sudo git clone https://github.com/iamtruptime/pythonapp.git
fatal: destination path 'pythonapp' already exists and is not an empty directory.
[ec2-user@python ~]$ ls
node-js-app-CICD pythonapp
[ec2-user@python ~]$ |
```

```
[ec2-user@python ~]$ cd pythonapp/
[ec2-user@python pythonapp]$ |
```

cd pythonapp

6.Create Virtual Environment

python3 -m venv myenv

```
[ec2-user@python pythonapp]$ sudo python3 -m venv myenv
[ec2-user@python pythonapp]$ |
```

7.Activate virtual environment

bash myenv/bin/activate

```
[ec2-user@python pythonapp]$ sudo python3 -m venv myenv
[ec2-user@python pythonapp]$ sudo bash myenv/bin/activate
[ec2-user@python pythonapp]$ |
```

8.Install dependencies

pip install -r requirements.txt

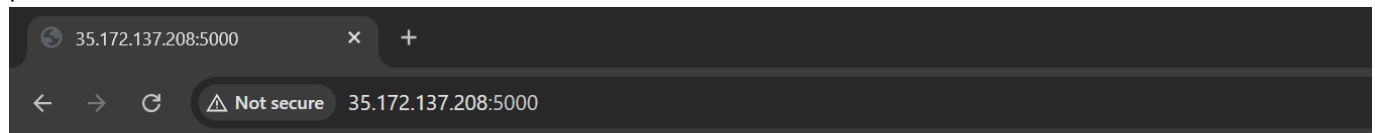
python3 app.py

```
ec2-user@python pythonapp]$ cd ..
ec2-user@python ~]$ cd pythonapp
ec2-user@python pythonapp]$ sudo pip install -r requirements.txt
Requirement already satisfied: click==8.0.3 in /usr/local/lib/python3.9/site-packages (from -r requirements.txt (line 1)) (8.0.3)
Requirement already satisfied: colorama==0.4.4 in /usr/lib/python3.9/site-packages (from -r requirements.txt (line 2)) (0.4.4)
Requirement already satisfied: Flask==2.0.2 in /usr/local/lib/python3.9/site-packages (from -r requirements.txt (line 3)) (2.0.2)
Requirement already satisfied: itsdangerous==2.0.1 in /usr/local/lib/python3.9/site-packages (from -r requirements.txt (line 4)) (2.0.1)
Requirement already satisfied: Jinja2==3.0.3 in /usr/local/lib/python3.9/site-packages (from -r requirements.txt (line 5)) (3.0.3)
Requirement already satisfied: MarkupSafe==2.0.1 in /usr/local/lib/python3.9/site-packages (from -r requirements.txt (line 6)) (2.0.1)
Requirement already satisfied: Werkzeug==2.0.2 in /usr/local/lib/python3.9/site-packages (from -r requirements.txt (line 7)) (2.0.2)
Requirement already satisfied: gunicorn==20.1.0 in /usr/local/lib/python3.9/site-packages (from -r requirements.txt (line 8)) (20.1.0)
Requirement already satisfied: setuptools>=3.0 in /usr/lib/python3.9/site-packages (from gunicorn==20.1.0->-r requirements.txt (line 8)) (59.6.0)
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended
that you create a virtual environment instead: https://pip.pypa.io/warnings/venv
ec2-user@python pythonapp]$ python app.py
bash: python: command not found
ec2-user@python pythonapp]$ sudo python3 app.py
* Serving Flask app 'app' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: on
* Running on all addresses.
  WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://172.31.46.88:5000/ (Press CTRL+C to quit)
* Restarting with stat
* Debugger is active!
* Debugger PIN: 460-718-499
```

9.Access the Application

public ip : port

port no -> 5000



successfully deployed python application through jenkins!!!!!!!!!!, added webhook

Key Features

- Deployment on AWS EC2
- Complete setup from scratch
- Virtual environment usage
- Dependency management with pip

Challenges Faced

- Connecting securely using SSH
- Managing Linux commands
- Setting up Python environment correctly

Project Summary

This project successfully demonstrates the deployment of a Python web application on a cloud platform. Starting from launching an EC2 instance, the process includes installing required tools, configuring the environment, and running the application. It provides practical experience in cloud computing, server management, and backend deployment, making it a strong foundation for real-world DevOps and backend development tasks.